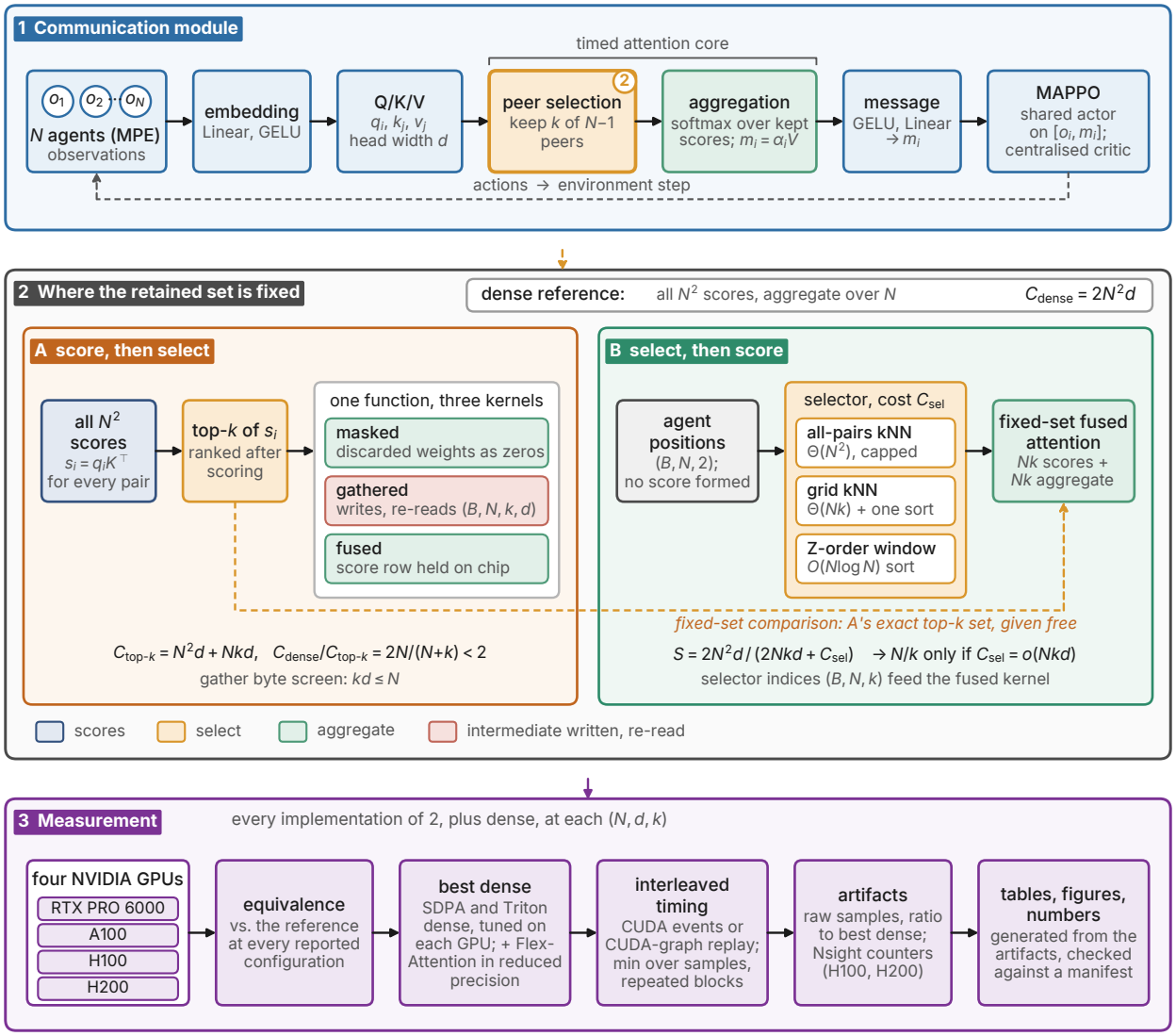

1 Communication module
timed attention core
o_1 o_2 ⋯ o_N
N agents (MPE)
observations
embedding
Linear, GELU
Q/K/V
q_i, k_j, v_j
head width d
peer selection
keep k of $N-1$ peers
2
aggregation
softmax over kept scores; $m_i = \alpha_i V$
message
GELU, Linear → m_i
MAPPO
shared actor on $[o_i, m_i]$; centralised critic
actions → environment step
2 Where the retained set is fixed
dense reference: all N^2 scores, aggregate over N $C_{dense} = 2N^2d$
A score, then select
all N^2 scores
$s_i = q_i K^\top$ for every pair
top-k of s_i
ranked after scoring
one function, three kernels
masked
discarded weights as zeros
gathered
writes, re-reads (B, N, k, d)
fused
score row held on chip
$C_{top\text{-}k} = N^2d + Nkd$, $C_{dense}/C_{top\text{-}k} = 2N/(N+k) < 2$
gather byte screen: $kd \le N$
B select, then score
agent positions
$(B, N, 2)$; no score formed
selector, cost C_{sel}
all-pairs kNN
$\Theta(N^2)$, capped
grid kNN
$\Theta(Nk)$ + one sort
Z-order window
$O(N \log N)$ sort
fixed-set fused attention
Nk scores + Nk aggregate
fixed-set comparison: A's exact top-k set, given free
$S = 2N^2d / (2Nkd + C_{sel})$ → N/k only if $C_{sel} = o(Nkd)$
selector indices (B, N, k) feed the fused kernel
scores
select
aggregate
intermediate written, re-read
3 Measurement
every implementation of 2, plus dense, at each (N, d, k)
four NVIDIA GPUs
RTX PRO 6000
A100
H100
H200
equivalence
vs. the reference at every reported configuration
best dense
SDPA and Triton dense, tuned on each GPU; + Flex-Attention in reduced precision
interleaved timing
CUDA events or CUDA-graph replay; min over samples, repeated blocks
artifacts
raw samples, ratio to best dense; Nsight counters (H100, H200)
tables, figures, numbers
generated from the artifacts, checked against a manifest